PROJECT : Corporate Training Resource & Lab Infrastructure Management Platform
Industry: Corporate Training / Education / L&D

Detailed Problem Statement

Organizations conducting technical training programs require classrooms, computer labs, trainers, software installations, systems, projectors, network facilities, and other resources. When multiple batches run simultaneously, manually allocating infrastructure can result in room conflicts, insufficient systems, missing software, and last-minute operational issues.

The organization requires a centralized platform that maintains courses, batches, trainers, classrooms, computer labs, software requirements, and infrastructure availability. Each training batch should have its infrastructure requirements defined before sessions begin.

The proposed **Corporate Training Resource & Lab Infrastructure Management Platform** should match batch requirements with available infrastructure. The system should detect room conflicts, capacity shortages, software incompatibility, and overlapping resource allocations.

Training managers should be able to monitor infrastructure utilization, upcoming requirements, allocation conflicts, and resource availability.

Functional Modules

- **Course Management**
  The system maintains courses, duration, technology requirements, and training specifications. Course information is used to identify resource requirements for batches.
- **Batch Management**
  The system maintains batch schedules, student count, course, location, and session dates. Batch information drives infrastructure allocation requirements.
- **Trainer Management**
  The system maintains trainer profiles, skills, availability, and assigned batches. Trainer information helps prevent scheduling conflicts.
- **Classroom Management**
  The system maintains classrooms with capacity, location, equipment, and availability. Classroom assignment must satisfy batch size and infrastructure requirements.
- **Computer Lab Management**
  The system maintains computer labs, system counts, specifications, and availability. Lab allocation is validated against batch size and technical requirements.
- **Software Requirement Management**
  Courses and batches can define required software, frameworks, versions, and configurations. The system verifies whether allocated labs satisfy the required software environment.
- **Resource Requirement Management**
  Each batch can specify infrastructure requirements such as systems, projector, internet, software, and classroom capacity. Requirements are compared against available resources before allocation.

- **Resource Allocation Management**
  Authorized users allocate classrooms, labs, trainers, and supporting resources to batches. Allocation records maintain start date, end date, resource, and batch information.

- **Conflict Detection Engine**
  The engine checks overlapping batch schedules, room allocations, lab availability, trainer assignments, capacity, and software requirements. Conflicts are displayed with specific reasons so they can be resolved before sessions begin.

- **Infrastructure Dashboard**
  Training managers can view resource utilization, upcoming batch requirements, allocation status, and unresolved conflicts. Dashboard information supports proactive infrastructure planning.

- **Maintenance & Unavailability Management**
  Classrooms, labs, or equipment can be marked unavailable for maintenance or other reasons. Existing allocations affected by such unavailability are flagged for reassignment.

- **Audit History**
  Resource allocations, changes, conflicts, and maintenance activities are recorded historically. Historical information supports infrastructure planning and accountability.

Core Business Engine

Training Infrastructure Allocation & Conflict Detection Engine